Predicting SpaceX Falcon 9 First-Stage Landing Success

Applied Data Science Capstone — data analysis report

Dataset supplied for this analysis: 101 launch records, dated 2010–2020, with launch site, booster, payload, orbit, customer, mission outcome and landing outcome.

Repository: https://github.com/sudharsan0317/spacex-capstone-project

Scope and honesty note: This report includes real exploratory plots and a small baseline model trained from the supplied CSV. It does not claim that the IBM course API extraction, web scraping, SQL lab, Folium map or Dash dashboard were completed. Add those outputs if the grading rubric explicitly requires them.

1. Executive Summary

Objective: investigate whether historical launch attributes can help classify whether a Falcon 9 first stage has a successful landing outcome.

The supplied CSV contains **101 rows**. Using a transparent label rule—landing outcome begins with “Success”—**61** records are labelled successful and **40** are not labelled successful.

A stratified 75/25 train/test split was used. Logistic Regression was the stronger of the two tested baseline models on this single split. The dataset is small, so the metrics are preliminary and should not be treated as reliable launch forecasts.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

2. Data and Target Definition

Fields used as predictors: payload mass, launch site, orbit, customer and booster family. Date was parsed to derive launch year for descriptive analysis, but launch year was not included in this baseline model.

Target definition: Landing_Success = 1 when Landing_Outcome starts with “Success”; otherwise 0. This groups “No attempt”, “Failure”, “Controlled (ocean)” and other non-success labels together, which is a simplifying project choice rather than a claim that all those outcomes mean the same operational thing.

Preprocessing: numeric payload mass was median-imputed and standardized; categorical variables were mode-imputed and one-hot encoded. Preprocessing was fitted within the training pipeline to reduce data leakage.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

3. Exploratory Analysis — Landing Outcomes

The chart shows the raw distribution of landing outcome categories. The outcome labels are not all equivalent: for example, “No attempt” is different from a failed landing. A binary target simplifies the modelling task, but this distinction should be acknowledged.

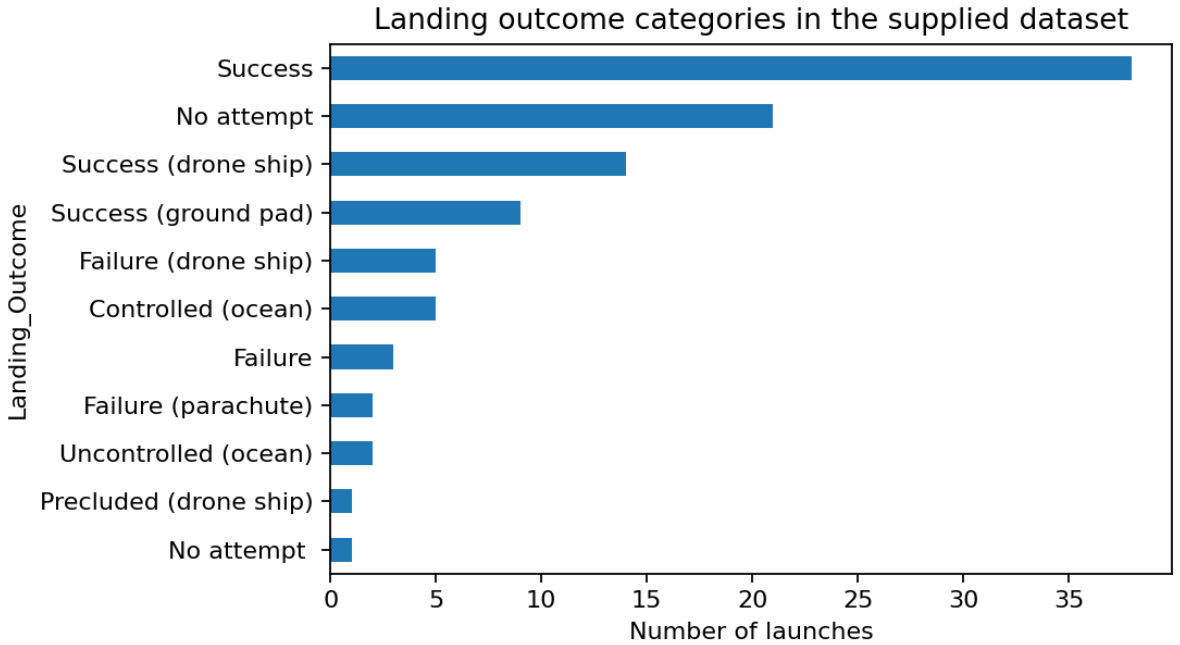


Figure 1. Raw landing outcome category counts from the provided CSV.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

4. Exploratory Analysis — Launch Sites

This chart counts records labelled successful by launch site. Counts depend on how many launches each site contributed; they are not success rates. A fair site comparison should also display the total launches per site and the corresponding percentage.

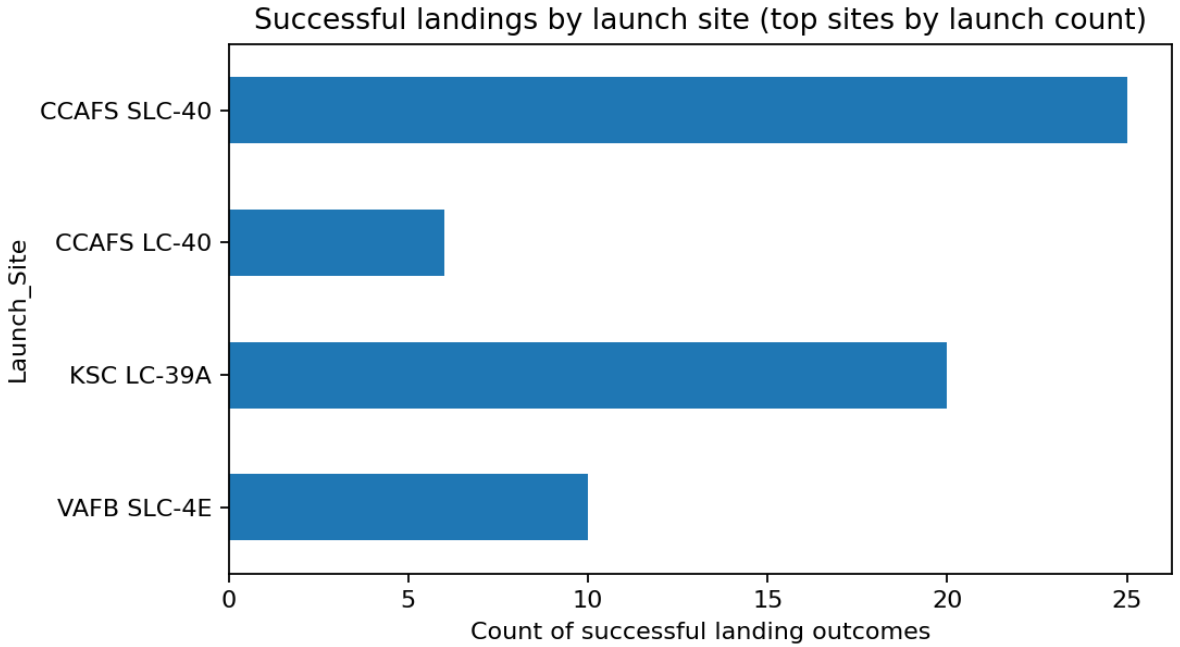


Figure 2. Successful-label counts by launch site among the most represented sites.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

5. Exploratory Analysis — Payload Mass

Payload mass is a potentially relevant predictor, but this scatter plot alone cannot establish a causal relationship. Some payload masses may be zero or missing-like values in the source data; inspect them before drawing strong conclusions.

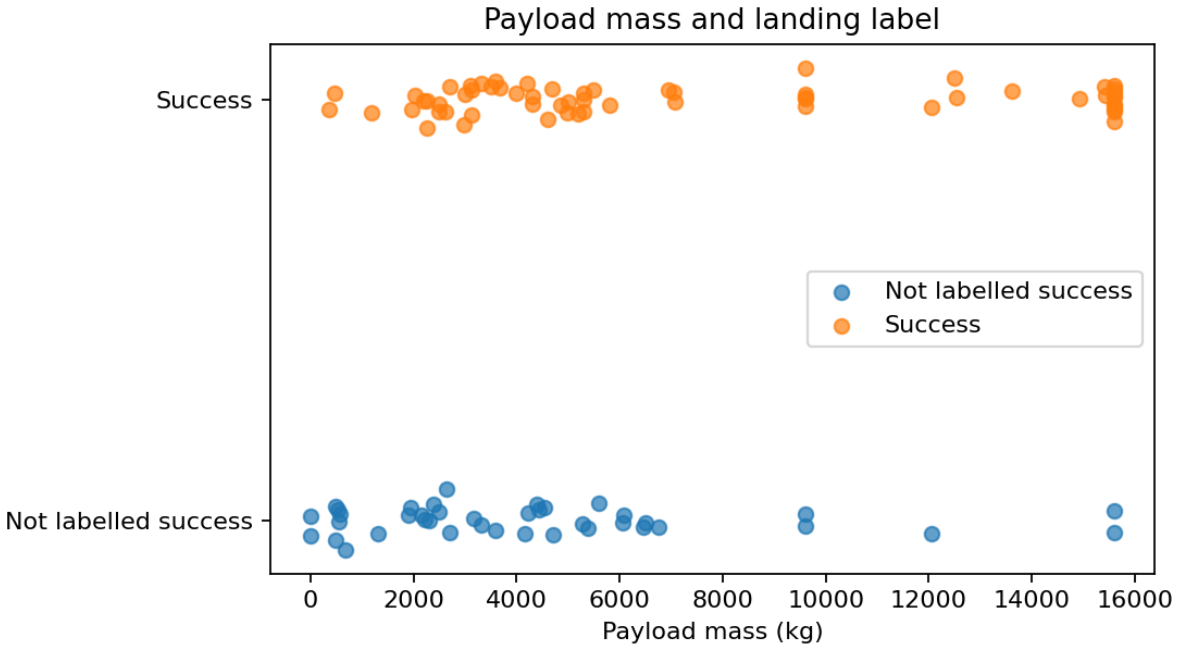


Figure 3. Payload mass against the binary landing label.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

6. Modelling Method

Two classifiers were compared: Logistic Regression and Random Forest. The same stratified train/test split and preprocessing approach were used for both models.

The dataset has only 101 rows. With a 25% test split, the test set has 26 records, so changing a few predictions can materially change the metrics. Cross-validation and more data would be needed for a more stable comparison.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

7. Model Comparison

Metrics on the single held-out test set:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.654	0.706	0.750	0.727
Random Forest	0.615	0.667	0.750	0.706

Logistic Regression achieved the higher accuracy and F1 score in this run. This does not prove it is universally the best model; performance can vary with the split and preprocessing choices.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

8. Confusion Matrix — Logistic Regression

The confusion matrix shows actual labels by predicted labels for the 26-record test set. Read this alongside precision and recall; accuracy alone can hide different types of error.



Figure 4. Confusion matrix for the Logistic Regression baseline.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

9. Results and Limitations

On this split, Logistic Regression: accuracy 0.654, precision 0.706, recall 0.750, F1 0.727.

Random Forest: accuracy 0.615, precision 0.667, recall 0.750, F1 0.706.

Limitations: only 101 records; one random train/test split; target collapses several distinct outcome types; no hyperparameter search; and no external validation. These results are educational and are not operational guidance.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv

10. Conclusion and Next Steps

This analysis demonstrates a compact workflow: inspect data, define a target, clean and encode features, explore distributions, train two classification baselines, and evaluate them on held-out records.

Next steps to fully align with the course rubric: include the course API extraction and web-scraping evidence; execute the SQL analysis and add query outputs; create the Folium launch-site map and Dash dashboard; compare models using cross-validation; and add screenshots or figures from those actual activities.

GitHub repository: https://github.com/sudharsan0317/spacex-capstone-project

Submission reminder: verify the rubric before uploading. This report does not substitute for any required notebook, map, dashboard, or GitHub deliverables.

IBM Applied Data Science Capstone • Analysis generated from supplied Spacex.csv